WORKED SOLUTION

STEP I 2019, Question 4

Pure • Factorising Quartic Equations

STEP I 2019, Question 4

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A quartic with no x^3 term has roots that add up to zero. So if we split the four roots into two pairs, the two pair-sums are negatives of each other, and that is exactly why the two quadratic factors have x-coefficients s and $-s$. Comparing coefficients then gives three equations, and eliminating p and q leaves a cubic in s^2. Part (i) is there to help with the surds at the very end.

Solution

Part (i)

Squaring $m+n\sqrt{2}$ gives $m^2+2n^2+2mn\sqrt{2}$, so we want $m^2+2n^2=3$ and $mn=1$. Taking $m=n=1$ works:

$$(1+\sqrt{2})^2 = 1+2\sqrt{2}+2 = 3+2\sqrt{2},$$

and since $1+\sqrt{2}>0$ it is the positive square root. Hence $m=1$, $n=1$:

$$\sqrt{3+2\sqrt{2}} = 1+\sqrt{2}.$$

In the same way, $(\sqrt{2}-1)^2 = 3-2\sqrt{2}$, so $\sqrt{3-2\sqrt{2}} = \sqrt{2}-1$. We shall need both.

Part (ii)

Why the factorisation exists. Let the four real roots be a, b, c, d, so $\mathrm{f}(x) = (x-a)(x-b)(x-c)(x-d)$. The coefficient of x^3 in f is $-(a+b+c+d)$, and it is zero, so $c+d=-(a+b)$. Grouping the roots in pairs,

$$\mathrm{f}(x) = (x^2-(a+b)x+ab)(x^2-(c+d)x+cd) = (x^2+sx+p)(x^2-sx+q),$$

with $s=-(a+b)$, $p=ab$ and $q=cd$, all real because the roots are real.

The three equations. Expanding,

$$(x^2+sx+p)(x^2-sx+q) = x^4+(p+q-s^2)x^2+s(q-p)x+pq.$$

Comparing with $x^4-10x^2+12x-2$:

$$\begin{aligned} p+q-s^2 &= -10,\\ s(q-p) &= 12,\\ pq &= -2. \end{aligned}$$

The equation for s. The second equation shows $s \neq 0$, so we may divide by it: $q-p = 12/s$. The first gives $p+q = s^2-10$. Now use the identity $(p+q)^2-(q-p)^2 = 4pq = -8$:

$$(s^2-10)^2 - \frac{144}{s^2} = -8.$$

Multiplying by s^2 (which is nonzero) gives

$$s^2(s^2-10)^2 + 8s^2 - 144 = 0,$$

as required.

The values of s^2. Put $u = s^2$. Expanding,

$$u(u-10)^2 + 8u - 144 = u^3 - 20u^2 + 108u - 144.$$

Trying small factors of 144, $u = 2$ gives $8-80+216-144 = 0$, so $u-2$ is a factor, and

$$u^3-20u^2+108u-144 = (u-2)(u^2-18u+72) = (u-2)(u-6)(u-12).$$

So the three possible values are $s^2 = 2$, 6 and 12.

> **Ask yourself** Why are there three values of s^2, when we only want one factorisation?

There are three ways to split four roots into two pairs, and each pairing gives its own pair of quadratic factors with $s^2 = (a+b)^2$. Any one of them will do for solving the equation; the smallest just gives the tidiest surds.

Solving $f(x) = 0$. Take $s^2 = 2$ and $s = \sqrt{2}$ (choosing $s = -\sqrt{2}$ simply swaps the two factors). Then

$$p+q = 2-10 = -8, \qquad q-p = \frac{12}{\sqrt{2}} = 6\sqrt{2},$$

so $p = -4-3\sqrt{2}$ and $q = -4+3\sqrt{2}$. (Check: $pq = 16-18 = -2$.) Hence

$$f(x) = (x^2+\sqrt{2}\,x-4-3\sqrt{2})(x^2-\sqrt{2}\,x-4+3\sqrt{2}).$$

For the first factor, the discriminant is

$$2+4(4+3\sqrt{2}) = 18+12\sqrt{2} = 6(3+2\sqrt{2}) = 6(1+\sqrt{2})^2,$$

using part (i), so its square root is $\sqrt{6}(1+\sqrt{2}) = \sqrt{6}+2\sqrt{3}$. The roots are

$$x = \frac{-\sqrt{2} \pm (\sqrt{6}+2\sqrt{3})}{2}.$$

For the second factor, the discriminant is

$$2 - 4(-4+3\sqrt{2}) = 18 - 12\sqrt{2} = 6(3-2\sqrt{2}) = 6(\sqrt{2}-1)^2,$$

with square root $\sqrt{6}(\sqrt{2}-1) = 2\sqrt{3}-\sqrt{6}$ (positive, since $2\sqrt{3} = \sqrt{12} > \sqrt{6}$). The roots are

$$x = \frac{\sqrt{2} \pm (2\sqrt{3}-\sqrt{6})}{2}.$$

So the four solutions of $f(x) = 0$ are

$$x = \tfrac{1}{2}(-\sqrt{2}+\sqrt{6}+2\sqrt{3}), \qquad x = \tfrac{1}{2}(-\sqrt{2}-\sqrt{6}-2\sqrt{3}),$$
$$x = \tfrac{1}{2}(\sqrt{2}+2\sqrt{3}-\sqrt{6}), \qquad x = \tfrac{1}{2}(\sqrt{2}-2\sqrt{3}+\sqrt{6}).$$

Checks

- Numerically the roots are about 2.2497, -3.6639, 1.2144 and 0.1998. They add to 0, as the missing x^3 term demands, and their product is about -2, the constant term.
- Substituting $x \approx 0.1998$: $x^4 - 10x^2 + 12x - 2 \approx 0.0016 - 0.3992 + 2.3976 - 2 \approx 0$.
- The sign changes of f agree: $f(0) = -2$, $f(1) = 1$, $f(2) = -2$, $f(3) = 25$, $f(-4) = 46$, so there is one root in each of $(-4,0)$, $(0,1)$, $(1,2)$ and $(2,3)$.

Common slips

- Not explaining why the x-coefficients are s and $-s$: it comes from the zero x^3 coefficient, i.e. the roots summing to zero.
- Forgetting to note $s \neq 0$ before dividing by s.
- Leaving the roots as $\sqrt{18 \pm 12\sqrt{2}}$: the question asks for full simplification, which is what part (i) is for.

Beyond STEP

This is Ferrari's method for the quartic in disguise: the cubic in s^2 is the *resolvent cubic*, and its three roots correspond to the three ways of pairing up the four roots. Why a quartic can be reduced to a cubic, while a general quintic cannot be solved by radicals at all, is the subject of Galois theory, which links the solvability of an equation to the symmetries of its roots. Allenby's *Rings, Fields and Groups* is a good first-year route towards it.

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